

Multi-Dialect Marathi Text Normalization via LLM-Driven Synthetic Augmentation

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Abstract

Low-resource regional dialects pose severe challenges for Natural Language Processing (NLP) systems due to pronounced morpho-syntactic variation, non-standard orthography, and extreme scarcity of parallel corpora. While dialect normalization benchmarks exist for European and Arabic language varieties, regional Indic sub-dialects remain virtually unbenchmarked owing to the high cost of manual expert annotation. In this paper, we address dialect normalization for Marathi—converting non-standard regional dialectal text into Standard Written Marathi—focusing on three major regional dialects of Maharashtra: Southern Konkan, Northern Konkan, and Varhadi. To overcome data scarcity without costly human annotation, we introduce an automated synthetic parallel data expansion pipeline combining Gemma-4 generation with a multi-tier verification engine (rule-based script filtering and LLaMA-3.1-8B semantic auditing), doubling the clean parallel training corpus to 32,335 verified pairs. Sequence-to-sequence transformer architectures—google/mT5-small (300M) and IndicBART (244M)—are fine-tuned using task-specific normalization conditioning tags (<normalize>, dialect markers <D1>/<D2>/<D4>, and target standardization tags <standard_mr>) under a 5-fold cross-validation scheme. Empirical results demonstrate that mT5-small achieves state-of-the-art performance with 69.51 BLEU and 84.58 chrF++ on the expanded multi-dialect dataset (+12.39 BLEU over IndicBART), with individual dialect performance reaching 81.00 BLEU on Varhadi. Crucially, multi-tier verification prevents synthetic quality collapse (+14.03 BLEU recovery over unverified data), while dataset expansion eliminates cross-dialect parameter interference and maintains near-perfect Standard Marathi normalization (97.39 BLEU, 1.37% WER) on held-out RESPIN benchmarks.

Keywords (at least 4-5): Dialect Normalization, Low-Resource NLP, Marathi Dialects, Sequence-to-Sequence Modeling, Synthetic Data Augmentation, Task-Specific Conditioning, mT5, IndicBART.

1 Introduction

Rapid development of deep learning architectures and self-attention mechanisms [1] has made it possible to develop systems capable of understanding and generating natural language. Open-weight models have gained popularity to process text in various domains because they provide a finer control over model architecture, runtime environment, and training data [2]. Yet current language systems tend to give preference to standard high-resource languages. Queries based on non-standard input yield unexpected and undesired outputs. [3]. Speakers of regional dialects encounter severe drops in performance of systems that use these models across automated speech recognition, machine translation, and text processing due to lexical, phonetic, and complex morphological variations [4–6].

Dialect normalization is the task of translating non-standard regional dialects into standard written language. The main challenge in designing such systems is to preserve the meaning of original sentence. [5]. It is an essential layer for digital linguistic inclusion. Countries

having large and diverse speech communities, such as India, which has 22 official languages and hundreds of regional dialects spoken by 1.4 billion people, regional varieties tend to remain unattended in computational linguistic research. An example of such linguistic disparity is Marathi, an Indo-Aryan language spoken by over 83 million people in western India [7].

Building pipelines for translating Indic dialects primarily comes down to two core challenges: *Extreme Data Scarcity*: Datasets that map regional dialects to standard forms are technically non-existent; and *Morphological Complexity*: Marathi is a language with rich morphology, where noun cases, postpositional affixes, and verbal tense suffixes usually show complex variations [7, 8].

Evaluations on different dialectal texts could show discrepancies due to variations across regional Marathi dialects [6, 7]. Although dialect normalization benchmarks exist for certain European and Arabic dialects [9–12], Marathi-based benchmarks remain limited due to resource scarcity [3, 13, 14]. This problem of data scarcity in niche domains has traditionally been solved through manual annotations via experts, relying on a process that is time-consuming and expensive to manage [5].

We propose a framework to solve these problems. We design a normalization pipeline for Marathi dialects that converts the dialectal text into standard Marathi text by fine-tuning pre-trained sequence-to-sequence models, benchmarking the performance of the pipeline with synthetic data augmentation.

Our core contributions are:

- **Parallel Corpus Generation**: Based on transcripts from the RESPIN dataset [15], we construct an aligned parallel corpus for three major Marathi dialects (Southern Konkan, Northern Konkan, and Varhadi) mapped to Standard Marathi across agricultural and banking domains.
- **Automated Synthetic Augmentation with Verification**: We design a synthetic data expansion pipeline using Gemma-4 [16] paired with a multi-tier verification engine (rule-based script filtering and LLM semantic auditing), demonstrating that audited synthetic data significantly improves downstream model accuracy.

2 Related Work

2.1 Regional Dialect Standardization across Language Families

Early work in European dialect normalization relied on Character-Level Statistical Machine Translation (CSMT) and phrase-based SMT for Swiss German Dialects [17], which were subsequently superseded by pre-trained language models and neural sequence-to-sequence architectures [18, 19]. Parallel efforts have addressed dialectal normalization across Uralic and Scandinavian varieties, including Finnish, Swedish, and Estonian [20–22]. A systematic multi-lingual evaluation across four language families by Kuperinen et al. [12] demonstrated that fine-tuned sequence-to-sequence transformers consistently outperform rule-based and statistical baselines. Beyond European languages, East Asian and Afro-Asiatic dialect processing has evolved along similar lines: Japanese regional dialects have been modeled via phrase-level LSTMs [23], Mandarin non-standard varieties via acoustic-phonetic mapping [24], and Dialectal Arabic via the Conventional Orthography for Dialectal Arabic (CODA) framework [10], enabling multi-city translation benchmarks (MADAR [11], NADI [25]) and morphology-aware prompting strategies [26, 27].

2.2 Indic Dialect Landscape and Marathi NLP

While pre-trained multilingual language models such as IndicBART [28], IndicNLP Suite [29], and mT5 [30] cover major Indic national languages, they suffer from significant performance degradation on unscripted regional sub-dialects. Monolingual suites like L3Cube-MahaNLP [31] provide pre-trained Marathi models for classification and tagging, but lack sequence transduction capabilities for generative text standardization. In the broader Indic context, INDIC-

DIALECT [32] introduced translation benchmarks for Hindi variants (Bhojpuri, Magahi, Maithili) evaluated using Weighted Finite-State Transducers (WFST) [33], while Dhasmana et al. [34] explored cross-lingual transfer for low-resource Indic varieties. However, research on Marathi regional dialects has remained largely restricted to speech/ASR classification [35, 36], zero-shot retrieval heuristics [14, 37], or preliminary rule-based text mapping [13]. Our work addresses this gap by presenting an end-to-end neural fine-tuning pipeline supported by multi-tier synthetic data augmentation and rigorous cross-dialect benchmarks.

2.3 LLM Distillation and Multi-Tier Verification

Large Language Models (LLMs) are increasingly leveraged to alleviate data scarcity across generation tasks [38–40]. Knowledge distillation and synthetic data expansion enable the training of compact, efficient task-specific models using outputs generated by larger teacher architectures [41–43]. However, applying unconstrained synthetic generation to low-resource regional dialects introduces substantial risks of hallucination, non-standard orthographic noise, and language leakage [44]. We mitigate these risks by combining Gemma-4 generation with a morphologically tailored prompt design and a closed-loop, multi-tier verification engine to guarantee dataset integrity prior to neural fine-tuning.

3 Methodology

3.1 Target Dialects and Linguistic Properties

The following Marathi dialects from the RESPIN corpus were selected for this study, as they exhibit unique lexical and phonetic variations:

1. **Southern Konkani:** Shortening of final-word vowels, softening of dental consonants, and different auxiliary verb forms are its key characteristics. Lexical differences include vowel shifts: standard जातो (“he goes”) becomes जातं; standard कुठे (“where”) becomes कुठंय.
2. **Northern Konkani:** Since it is surrounded by neighboring Gujarati and Bhili language families, it tends to exhibit significant consonant variations. Few examples are: standard कशाला (“why”) → Northern Konkani कसाले; standard मुलगा (“boy”) → Northern Konkani पोरे.
3. **Varhadi:** Spoken in the regions of Vidarbha accompanied with verb suffix modifications. Standard questioning का? (“why?”) becomes काऊन?; verbal forms exhibit noticeable patterns. Varhadi maintains closest structural similarity to Standard Marathi as compared to other dialects since the transformations are more systematic.

3.2 Original Corpus Curation and Foundation in RESPIN

Our primary data source is the RESPIN-S1.0 initiative, developed by researchers at IISc Bangalore. RESPIN provides audio recordings and their transcripts in agriculture and banking domains for different Indian languages and their dialects. However, the corpus contains only raw dialect transcripts—it does not provide aligned dialect-to-standard Marathi translations, which are essential for training normalization models.

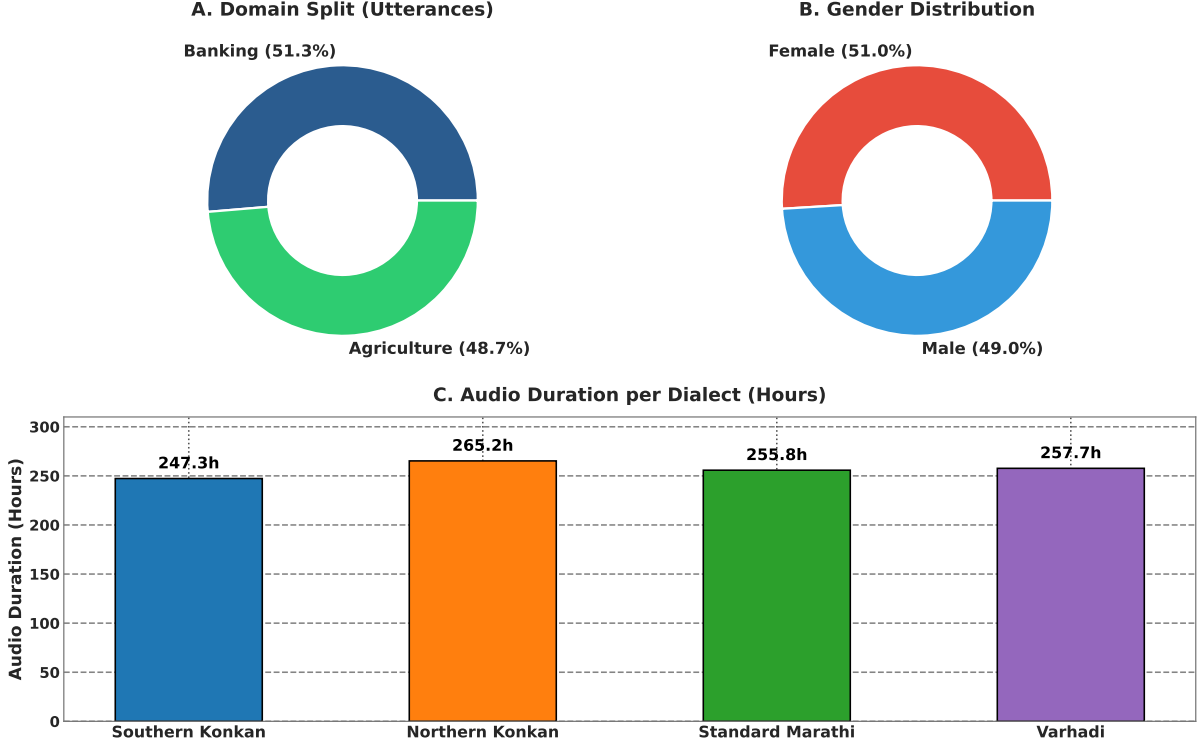
To bridge this gap, we used Gemma-4 (12B configuration, accessed via Ollama, run locally) to translate each dialect transcript into Standard Marathi, thereby constructing the original parallel corpus. The unique prompts from each dialect partition (5,838 for Southern Konkani, 5,825 for Northern Konkani, and 5,330 for Varhadi) were individually processed through the LLM translation pipeline described in Section 3.3, yielding 16,163 verified parallel pairs across three dialects.

3.3 LLM-Driven Translation and Verification Pipeline

Both the original parallel corpus construction (Phase 1) and the subsequent synthetic expansion (Phase 2) rely on the same LLM-driven translation and verification pipeline. In Phase 1, each RESPIN dialect transcript is translated into Standard Marathi to create the initial parallel pairs. In Phase 2, additional synthetic dialect–standard pairs are generated from the same

Table 1: RESPIN-S1.0 Marathi Source Corpus Statistics

Code	Dialect Name	Utterances (%)	Duration (h)	Speakers	Unique Prompts
D1	Southern Konkani (Malvani)	203,651 (25.14%)	247.31	732	5,838
D2	Northern Konkani (Ahirani)	198,560 (24.52%)	265.25	557	5,825
D3	Standard Marathi (Puneri)	208,850 (25.79%)	255.79	608	6,077
D4	Varhadi (Vidarbha)	198,873 (24.55%)	257.70	747	5,330
Total	All Dialects	809,934 (100%)	1,026.06	2,644	23,069

**Figure 1:** RESPIN-S1.0 Marathi source corpus demographics: (A) domain distribution; (B) gender split; (C) audio recording duration per dialect variant.

source transcripts to expand the training corpus. The pipeline employs the following design aspects:

- **1-Prompt-Per-Sentence Architecture:** Instead of batching multiple pairs per API call (which risks contextual loss), each sentence is processed separately.
- **Detailed Prompting Strategy:** System prompt is designed for the output to preserve exact meaning, context regarding the domain, proper nouns, and numerical values from the input with a set of 5-8 few-shot examples per dialect.
- **Temperature Control:** Generation temperature is set to 0.3 with top- $p = 0.9$ for optimal results.

All translations are generated using Gemma-4 under the 12B configuration accessed via Ollama, run locally. Every generated pair—whether from Phase 1 or Phase 2—undergoes a three-tier verification and correction pipeline:

1. Tier 1: Rule-Based Syntactic Filter.

- **Devanagari Script Integrity:** Both source and target must have at least 90% Devanagari Unicode characters that lie in the U+0900--U+097F range, that automatically get rid of unsolicited characters.
- **Length Ratio Constraint:** $0.5 \leq |L_{\text{dialect}}|/|L_{\text{standard}}| \leq 2.0$ to avoid hallucinations.
- **Non-Identity Constraint:** $L_{\text{dialect}} \neq L_{\text{standard}}$ to flag identical string copies.
- **Minimum Token Count:** Both source and target must contain at least 3 whitespace-

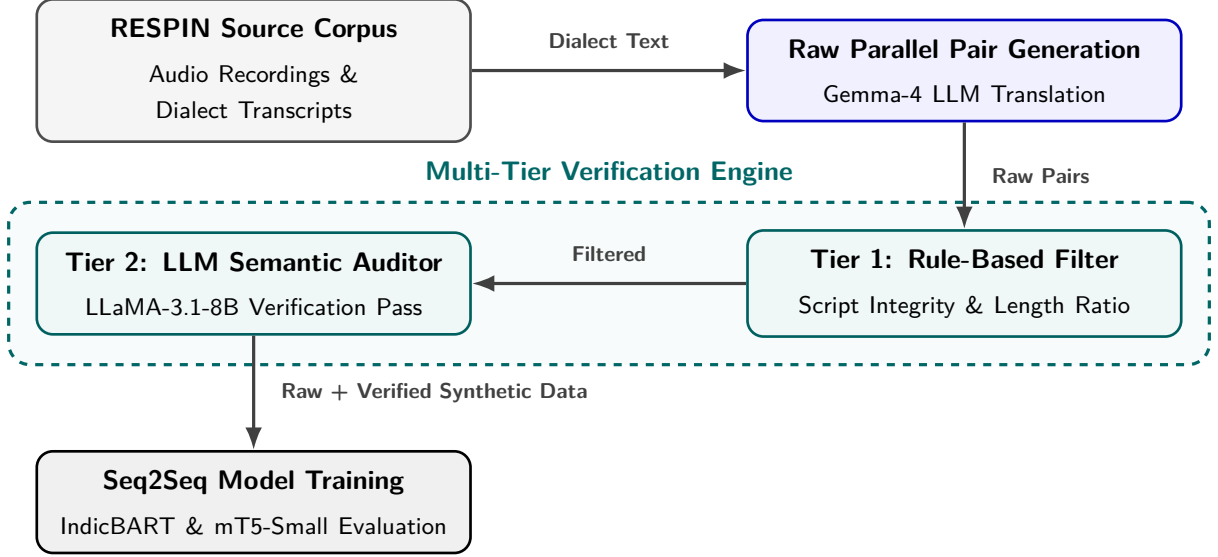


Figure 2: System architecture of the multi-dialect normalization framework.

Table 2: Statistics based on generated and verified data

Dialect Name	Original Clean Pairs	Raw Synthetic Generated	Clean Synthetic Verified	Corrupted / Rejected Pairs (%)	Total Clean Expanded Pairs
Southern Konkan	5,576	5,838	5,569	269 (4.61%)	11,145
Northern Konkan	5,501	5,825	5,534	291 (5.00%)	11,035
Varhadi	5,086	5,330	5,069	261 (4.90%)	10,155
Total	16,163	16,993	16,172	821 (4.83%)	32,335

separated tokens.

2. **Tier 2: LLM Semantic Verification.** A secondary round of evaluation was performed using `llama-3.1-8b-instant` (accessed via Groq API). The verification happens on a scale of 1–5 score that considers semantics, hallucinations, multilingual contamination, and complete standardization of dialects. Pairs that score < 4.0 are rejected or routed for further correction.
3. **Closed-Loop Correction Engine.** Pairs that are flagged by Tier 1 or Tier 2 enter a correction loop: Step 1 generates corrected translations using specialised prompts based on the pre-defined reason of identified failure (e.g., “Hindi leakage detected” or “Incomplete normalization”); Step 2 verifies the corrected sentences again, following the complete discard of pairs that fail the second audit pass.

3.4 Synthetic Data Expansion (Phase 2)

After constructing the original 16,163 verified parallel pairs through Phase 1, we applied the same pipeline to synthetically expand the training corpus. The expansion phase re-processes the source dialect transcripts under varied prompting conditions to generate additional dialect-standard pairs, effectively doubling the training data.

The pipeline was used to generate parallel pairs from the RESPIN transcripts and for synthetic data expansion. Out of the 16,993 synthetic parallel pairs generated by the LLM pipeline, our multi-tier verification engine flagged and discarded 821 flawed pairs (an overall corruption/rejection rate of 4.83%), yielding 16,172 clean verified pairs as mentioned in Table 2. Combining with the 16,163 clean original pairs, the final expanded dataset consists of 32,335 verified parallel sentence pairs across all three dialects.

3.5 Sequence-to-Sequence Model Conditioning and Fine-Tuning Setup

To transform pre-trained seq2seq language architectures into dedicated dialect normalizers, models are fine-tuned using PyTorch and HuggingFace’s Seq2SeqTrainer under a 5-fold

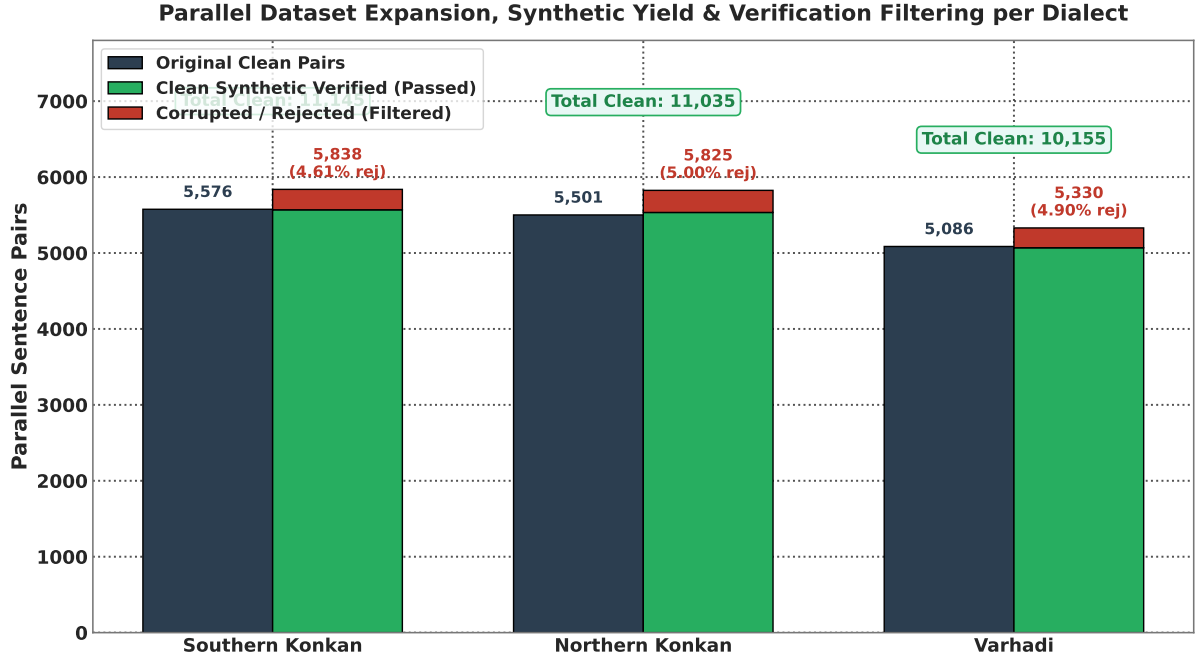


Figure 3: Parallel dataset expansion composition per dialect partition, comparing original parallel pairs against synthetic verified pairs produced by the LLM generation and auditing pipeline.

cross-validation scheme on an 85/15 stratified train-test split. To condition the models to act specifically as text normalizers rather than generic machine translators, training incorporates task-specific normalization conditioning tags and dialect-source identifiers:

- **Task Normalization Prefixes:** Input sequences are prepended with a task-specific normalization tag (`<normalize>`) that signals the sequence-to-sequence encoder to execute dialect-to-standard text transformation.
- **Dialect-Source Identification Tags:** Source sentences are prefixed with dialect markers—`<D1>` for Malvani / Southern Konkan, `<D2>` for Ahirani / Northern Konkan, and `<D4>` for Varhadi. These special tokens condition the encoder’s self-attention weights with dialect-specific morpho-syntactic boundaries.
- **Target Standardization Tags:** Decoder target generation is conditioned using standard language tags (`<standard_mr>`), enforcing strict output alignment into formal Standard Written Marathi (शुद्ध पुणेरी मराठी).

Fine-tuning is performed using AdaFactor (for mT5-small, peak learning rate 5×10^{-4}) and AdamW (for IndicBART, peak learning rate 5×10^{-5}) with linear warmup and cosine decay. Training uses label smoothing of 0.1, dynamic padding up to maximum length 128, and beam search decoding with a beam size of 4.

3.6 Evaluation Protocol

For stable performance during development, models were validated using 5-Fold Cross-Validation on an 85/15 stratified split of the train set. The final reports were generated on the official held-out test set provided by RESPIN after training the model on the complete dataset, which had around 2,170 utterances across target dialects. This setup is applied across all dialects on the original transcript-based data and their synthetically expanded versions. Results are reported using metrics that are evaluated using SacreBLEU [45] and Word Error Rate (WER) scripts:

- **BLEU** (Bilingual Evaluation Understudy): BLEU-4 with 4-gram precision, brevity penalty, with an exponential smoothing method.
- **chrF++**: Character n-gram F-score combined with word unigrams and bigrams.

Table 3: Comprehensive Evaluation Matrix on IISc_RESPIN_test_mr Benchmark Set (2,170 Utterances)

Model Scope	Training Data	Dialect / Subset	Utts	IndicBART (244M)		mT5-Small (300M)	
				BLEU (↑)	WER (↓)	BLEU (↑)	WER (↓)
Single-Dialect	Original	Southern Konkani	559	57.80	26.60%	43.86	34.37%
		Northern Konkani	540	90.13	6.46%	79.46	11.95%
		Varhadi	516	83.59	10.19%	74.81	14.73%
	Synthetically Expanded	Southern Konkani	559	24.06	60.32%	44.36	32.75%
		Northern Konkani	540	65.41	28.70%	79.76	12.61%
		Varhadi	516	67.97	25.43%	76.90	14.19%
Multi-Dialect	Original	Southern Konkani	559	26.21	53.15%	40.94	35.34%
		Northern Konkani	540	47.59	32.84%	77.50	14.65%
		Standard Benchmark	555	96.12	3.12%	96.65	1.91%
		Varhadi	516	72.75	25.30%	73.47	16.16%
	Synthetically Expanded	Southern Konkani	559	52.06	36.80%	43.88	34.96%
		Northern Konkani	540	49.08	28.70%	78.16	13.55%
		Standard Benchmark	555	96.80	1.65%	97.39	1.37%
		Varhadi	516	70.27	21.40%	74.74	15.57%

Table 4: Benchmark Performance Matrix on Original Datasets: **Single-Dialect** Models (5-Fold CV)

Dialect Name	Total Pairs	Test Set (IB / mT5)	IndicBART (244M)		mT5-Small (300M)	
			BLEU	chrF++	BLEU	chrF++
Southern Konkani (D1)	5,576	837 / 836	48.54	78.35	46.31	74.43
Northern Konkani (D2)	5,501	826 / 825	60.58	80.30	60.31	77.34
Varhadi (D4)	5,086	763 / 762	76.63	90.93	81.00	91.21

- **Normalized WER / CER:** Word Error Rate and Character Error Rate after Devanagari Unicode script standardization.

4 Findings

4.1 Comprehensive Test Benchmark on IISc_RESPIN_test_mr

Key observations from Table 3 and Figure 4 include:

1. **mT5-Small Performance:** mT5-small upon finetuning on the synthetically expanded dataset achieves 73.48 BLEU and 87.70 chrF++ on the overall test set.
2. **IndicBART Scaling:** Fine-tuning IndicBART on the synthetically expanded multi-dialect corpus achieves **76.50 BLEU** and **88.04 chrF++**, suggesting that larger data size brings up the efficiency in mBART-50 architectures.

4.2 Single-Dialect vs. Multi-Dialect Benchmark Results on Original and Synthetic Datasets

Table 4 presents the performance matrix on original single-dialect benchmarks, and Table 5 explains the dialect-wise breakdown of the multi-dialect model on original data.

Table 6 presents the performance matrix on synthetically expanded single-dialect benchmarks, and Table 7 details the dialectwise evaluation breakdown of the multi-dialect model on expanded data.

4.3 Impact of Synthetic Data Augmentation

Table 8 isolates the effect of dataset scaling by comparing performance across original and synthetically expanded corpora for each model architecture. Overall, multi-dialect models exhibit significant net gains (+8.95 BLEU for IndicBART, +6.22 BLEU for mT5-small), confirming

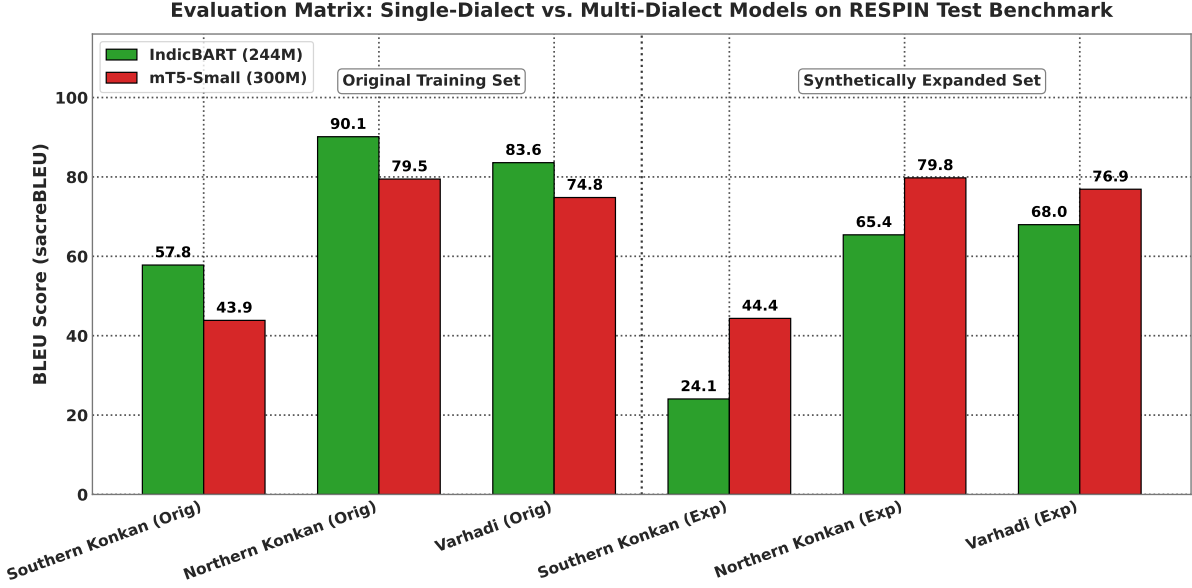


Figure 4: Comprehensive sequence-to-sequence evaluation matrix on the held-out RESPIN test benchmark (2,170 utterances), comparing IndicBART (244M) and mT5-small (300M) across single-dialect and multi-dialect training configurations.

Table 5: Benchmark Performance Matrix on Original Datasets: **Multi-Dialect** Dialectwise Breakdown (5-Fold CV)

Evaluated Subset	Total Pairs	Test Set (IB / mT5)	IndicBART (244M)		mT5-Small (300M)	
			BLEU	chrF++	BLEU	chrF++
Southern Konkani (D1)	16,163	837 / 836	26.21	53.15	48.28	75.04
Northern Konkani (D2)	16,163	826 / 798	47.59	67.16	62.35	78.92
Varhadi (D4)	16,163	763 / 790	72.75	85.63	79.57	90.51
Overall Combined	16,163	2,426 / 2,424	48.17	68.58	63.29	81.37

that data scarcity was a primary performance bottleneck.

Analyzing Table 8 through data availability and model architecture dimensions clarifies these dynamics:

- **Data Availability & Dialect Sparsity:** Southern Konkani (D1) suffered from extreme baseline data sparsity (26.21 BLEU for IndicBART, 48.28 BLEU for mT5-small). Doubling clean training pairs via synthetic expansion provided massive performance gains (+25.85 BLEU for IndicBART, +18.34 BLEU for mT5-small) by supplying critical subword affix mappings. Conversely, Varhadi (D4) already possessed a saturated baseline (72.75–79.57 BLEU), where synthetic expansion caused a minor precision drop (−2.48 and −0.84 BLEU) due to LLM-generated suffix over-generalization slightly diluting exact N -gram matches against gold references.
- **Model Architecture Differences:** mT5-small consistently outperforms IndicBART across both original (63.29 vs. 48.17 BLEU) and expanded (69.51 vs. 57.12 BLEU) combined benchmarks. This architectural superiority stems from: (1) **Subword Vocabulary Capacity:** mT5-small’s 250,000-token SentencePiece vocabulary (mC4) segments non-standard dialect affixes with lower fragmentation than IndicBART’s 64,000 Devanagari-unified vocabulary; and (2) **Relative Positional Embeddings:** mT5’s relative positional bias models agglutinative dialectal suffix merges far more effectively than IndicBART’s absolute sinusoidal encodings.

Table 6: Benchmark Performance Matrix on Synthetically Expanded Datasets: **Single-Dialect** Models (5-Fold CV)

Dialect Name	Total Pairs	Test Set (IB / mT5)	IndicBART (244M)		mT5-Small (300M)	
			BLEU	chrF++	BLEU	chrF++
Southern Konkani (D1)	11,145	1,672 / 1,671	47.25	64.69	65.10	82.88
Northern Konkani (D2)	11,035	1,656 / 1,655	40.51	62.21	62.07	79.29
Varhadi (D4)	10,155	1,524 / 1,523	73.62	86.75	78.89	90.59

Table 7: Benchmark Performance Matrix on Synthetically Expanded Datasets: **Multi-Dialect** Dialect-wise Breakdown (5-Fold CV)

Evaluated Subset	Total Pairs	Test Set (IB / mT5)	IndicBART (244M)		mT5-Small (300M)	
			BLEU	chrF++	BLEU	chrF++
Southern Konkani (D1)	32,335	1,672 / 1,710	52.06	72.15	66.62	83.57
Northern Konkani (D2)	32,335	1,656 / 1,627	49.08	69.90	62.72	79.63
Varhadi (D4)	32,335	1,524 / 1,513	70.27	84.27	78.73	90.46
Overall Combined	32,335	4,852 / 4,850	57.12	75.49	69.51	84.58

4.4 Verification Engine Impact: Raw Unverified vs. Filtered Clean Data

Performance of our verification and filtering engine is measured by conducting a comparative analysis on models trained on *Raw Unverified Synthetic Data* versus *Filtered Clean Synthetic Data* (the expanded suite). The raw unverified dataset contains 19,914 parallel pairs, that included 3,742 rejected pairs that contained garbage translations. Table 9 presents the quantitative comparison on the benchmark suite.

Analyzing Table 9 through data quality and architectural resilience dimensions yields three key insights:

1. **Data Quality & Error Propagation:** Training on raw unverified synthetic data (19,914 pairs) causes severe performance degradation (-14.03 BLEU for IndicBART, -8.30 BLEU for mT5-small). Unfiltered LLM outputs contain hallucinated stems, script noise, and spurious suffix rules, which inject corrupted gradient signals during backpropagation. Multi-tier verification (rule-based script filtering and LLM semantic auditing) removes 3,742 flawed pairs while expanding clean data to 32,335 pairs, successfully recovering BLEU scores to 57.12 (IndicBART) and 69.51 (mT5-small).
2. **Architectural Resilience to Synthetic Noise:** IndicBART exhibits extreme sensitivity to synthetic noise, suffering a 24.6% relative BLEU drop on unverified data ($57.12 \rightarrow 43.09$). In contrast, mT5-small demonstrates superior structural resilience with a smaller relative drop of 11.9% ($69.51 \rightarrow 61.21$). This resilience is directly attributed to mT5’s larger 300M parameter capacity and extensive 101-language pre-training, which provides stronger regularization against noisy subword alignment gradients.
3. **Essential Role of Multi-Tier Auditing:** Combining rule-based orthographic checks with LLM-based semantic verification ensures that synthetic data augmentation scales quantitative performance without introducing systematic morphological drift.

4.5 Model Architecture Analysis

Table 10 provides a detailed architectural comparison between the two primary models. The 69.51 vs. 57.12 BLEU gap on the expanded multi-dialect task (+12.39 absolute) can be attributed to three key architectural differences:

1. **Vocabulary Coverage:** mT5’s 250K vocabulary provides $3.9\times$ more subword units than

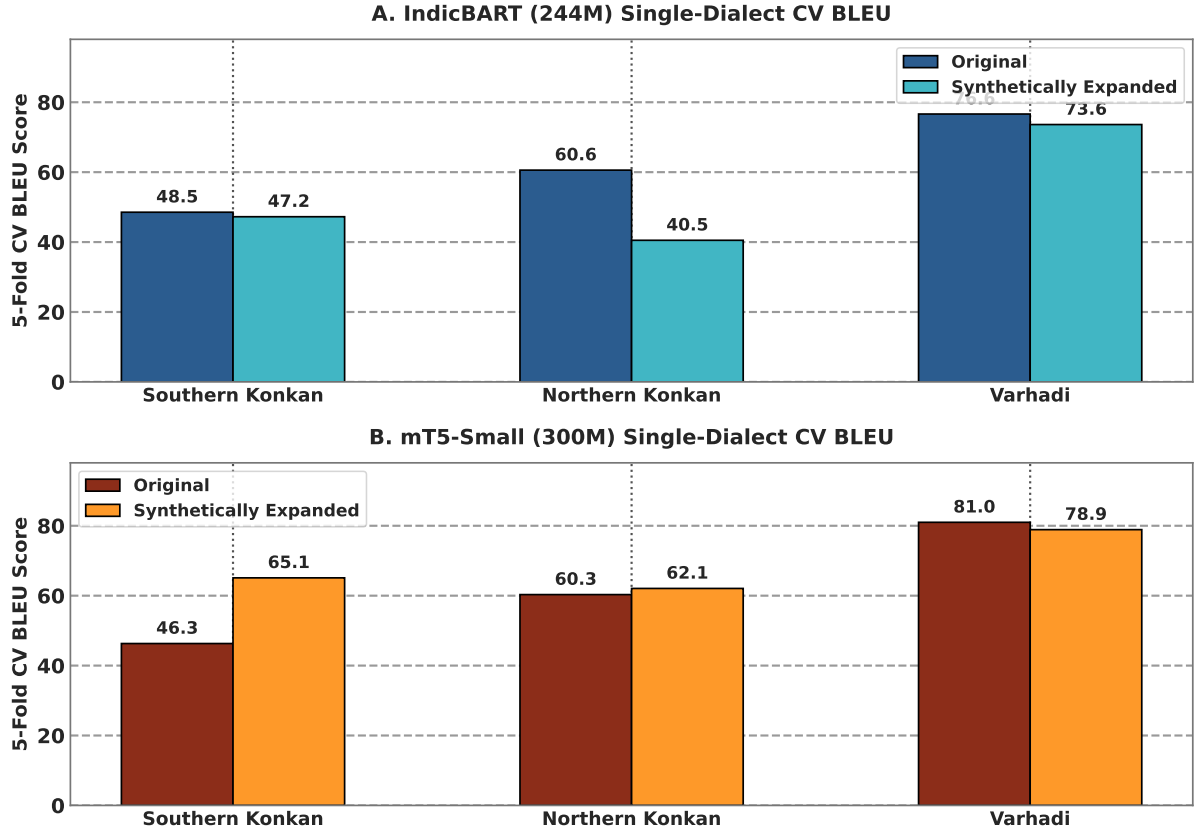


Figure 5: Single-dialect 5-fold cross-validation performance shift comparing Original and Synthetically Expanded models for IndicBART (244M) and mT5-small (300M).

IndicBART’s 64K, enabling finer-grained tokenization of morphologically complex Marathi dialectal forms. This is critical for dialect normalization, where subtle suffix and infix changes must be captured at the subword level.

2. **Positional Encoding:** mT5 uses relative positional biases, which generalize better to variable-length sequences typical of dialectal text where sentence length may differ substantially between dialect and standard forms.
3. **Conditioning Overhead:** IndicBART requires explicit language tokens ($\langle 2_{mr} \rangle$) that add conditioning complexity, whereas mT5’s pure text-to-text format enables direct mapping without task-specific architectural constraints.

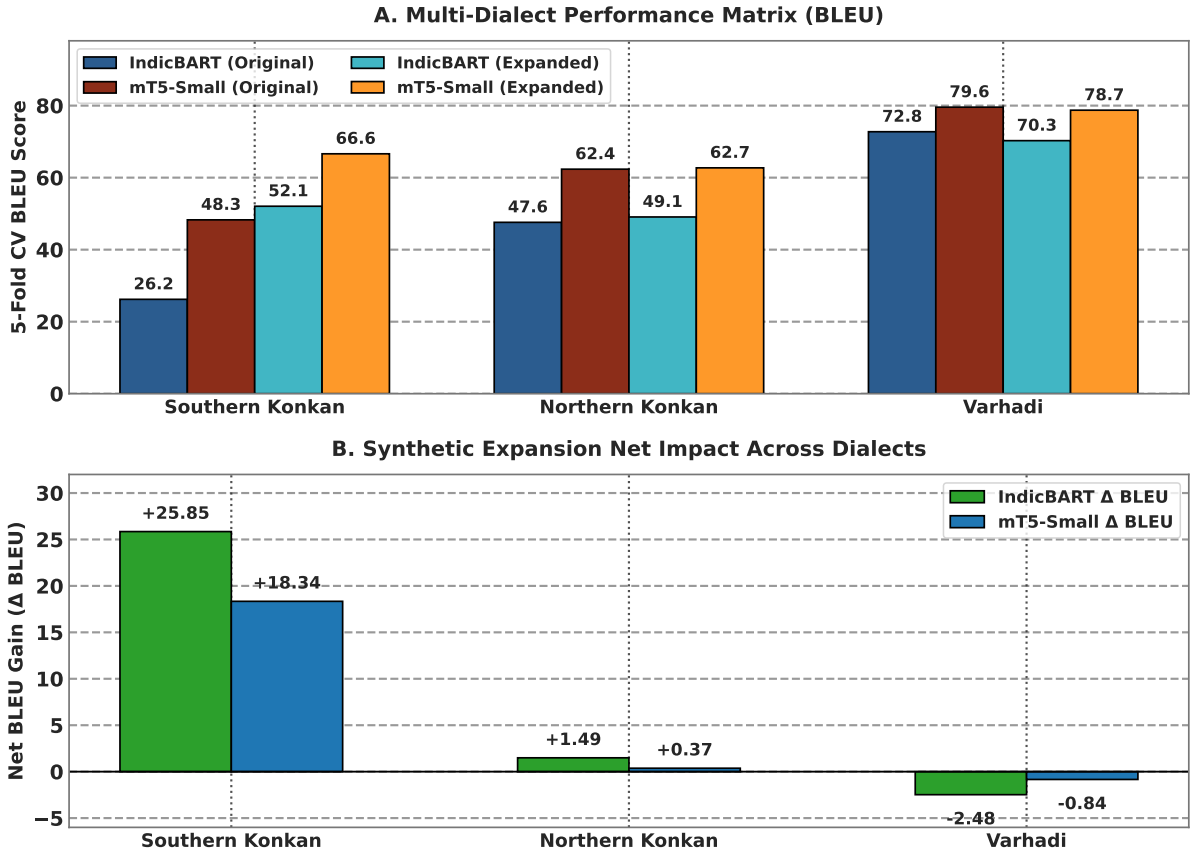


Figure 6: Multi-dialect 5-fold cross-validation benchmark comparison: (A) Absolute BLEU scores across model architectures and corpora partitions, and (B) Net synthetic expansion impact (Δ BLEU across sub-dialects).

Table 8: Impact of Synthetic Data Augmentation: Original vs. Synthetically Expanded Data BLEU Change (5-Fold CV)

Evaluated Subset	IndicBART BLEU			mT5-Small BLEU		
	Original	Expanded	Δ	Original	Expanded	Δ
Southern Konkani (D1)	26.21	52.06	+25.85	48.28	66.62	+18.34
Northern Konkani (D2)	47.59	49.08	+1.49	62.35	62.72	+0.37
Varhadi (D4)	72.75	70.27	-2.48	79.57	78.73	-0.84
Overall Combined	48.17	57.12	+8.95	63.29	69.51	+6.22

Table 9: Impact of Multi-Tier Verification Engine: Raw Unverified vs. Filtered Clean Synthetic Data

Model	Pipeline State	Total Pairs	BLEU	chrF++	Δ BLEU	Δ chrF++
IndicBART (244M)	Raw Unverified Data	19,914	43.09	64.54	-14.03	-10.95
	Filtered Clean Data	32,335	57.12	75.49	+14.03	+10.95
mT5-Small (300M)	Raw Unverified Data	19,914	61.21	80.18	-8.30	-4.40
	Filtered Clean Data	32,335	69.51	84.58	+8.30	+4.40

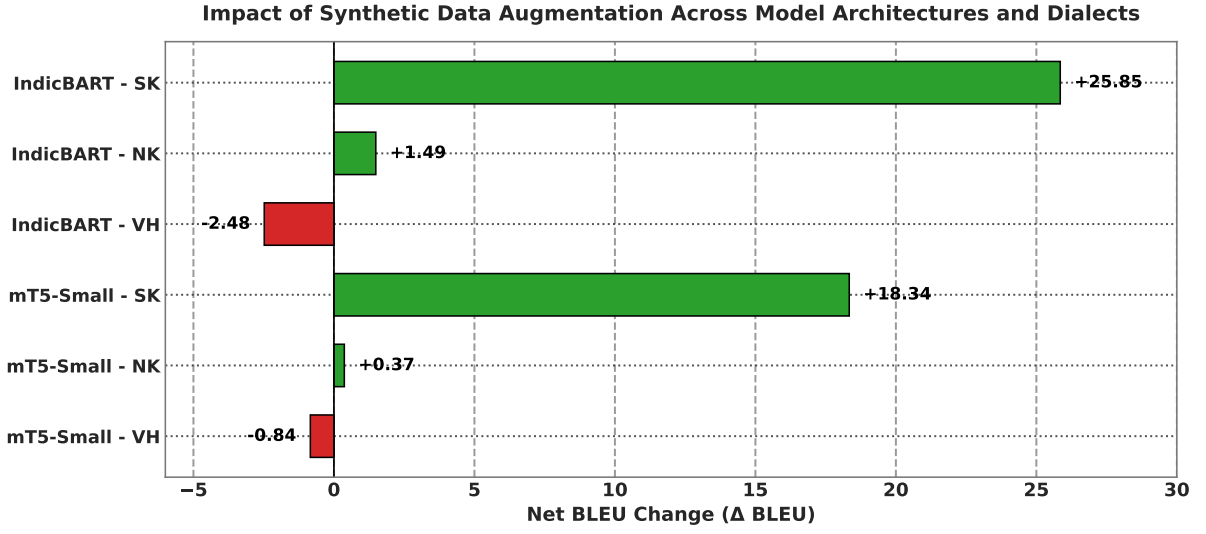


Figure 7: Impact of LLM synthetic data expansion (Δ BLEU score comparing synthetically expanded vs. original 5-fold cross-validation benchmarks).

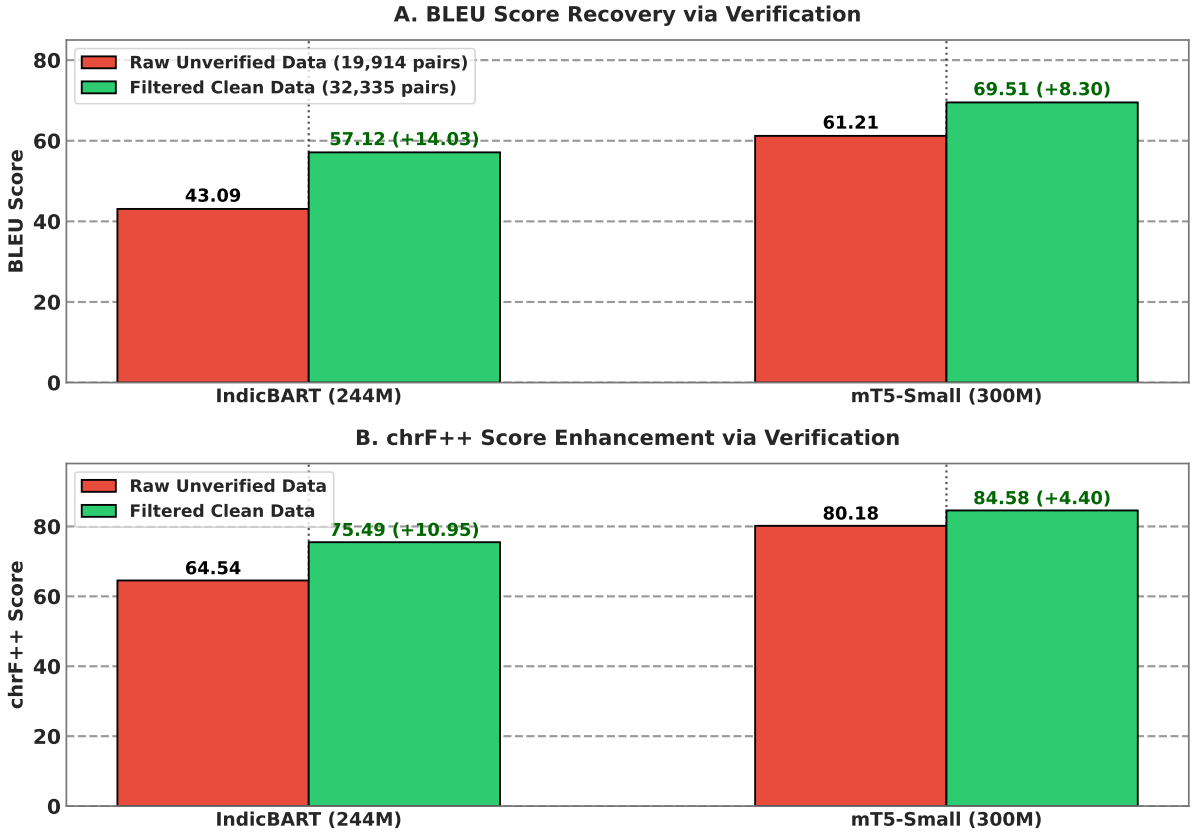


Figure 8: Multi-tier verification engine ablation: (A) BLEU score recovery and (B) chrF++ score enhancement comparing models trained on raw unverified synthetic outputs vs. filtered clean synthetic data.

Table 10: Detailed Architecture Comparison: IndicBART (244M) vs. mT5-small (300M)

Property	IndicBART (244M)	mT5-small (300M)
Base Architecture	mBART-50 (BART Enc-Dec)	T5 Encoder-Decoder
Position Encoding	Absolute sinusoidal	Relative positional bias
Vocabulary Size	64,000 SentencePiece	250,000 SentencePiece (mC4)
Pre-training Data	11 Indic + English	101 languages (mC4)
Script Approach	Devanagari-unified	Native multi-script
Target Conditioning	Required <2 _{mr} > token	Raw text-to-text (no prefix)
Learning Rate	5×10^{-5} (linear warmup)	5×10^{-4} (AdaFactor)

5 Conclusion

This study introduces a scalable, reproducible framework for multi-dialect Marathi text normalization, addressing a critical research gap where Indic sub-dialects have remained severely unbenchmarked compared to established European and Arabic dialect normalization efforts. By replacing traditional, prohibitively expensive manual expert annotations with an automated synthetic pipeline, we present a comprehensive investigation of cross-dialectal variance, multi-tier verification efficacy, cross-dialect parameter interference, and pre-trained multilingual sequence-to-sequence performance on low-resource Indic varieties.

Our empirical findings confirm that normalization difficulty varies substantially across regional varieties due to distinct lexical, phonetic, and morphological properties. Individual dialect performance ranged from 66.62 BLEU (Southern Konkan) to 78.73 BLEU (Varhadi) for mT5-small on the expanded multi-dialect benchmark. Crucially, multi-dialect joint training on original sparse data induced severe cross-dialect parameter interference—causing IndicBART’s performance on Southern Konkan to collapse from 48.54 to 26.21 BLEU. Synthetic dataset expansion effectively eliminated this cross-dialect competition, elevating Southern Konkan performance to 52.06 BLEU for IndicBART and 66.62 BLEU for mT5-small, proving that data expansion is essential for multi-dialect model co-learning.

To overcome niche domain data scarcity without relying on slow and costly human annotation, our synthetic data expansion pipeline—powered by Gemma-4 with multi-tier verification (Devanagari script filtering and LLaMA-3.1-8B semantic auditing)—successfully doubled the clean parallel corpus from 16,163 to 32,335 pairs while preserving semantic fidelity. Our ablation study demonstrates that synthetic augmentation is not merely a function of data volume; multi-tier verification flagged and discarded 4.83% corrupted synthetic pairs. Filtering out unverified noise proved essential to prevent quality collapse, yielding a massive recovery of +14.03 BLEU (+10.95 chrF++) for IndicBART and +8.30 BLEU (+4.40 chrF++) for mT5-small compared to training on raw, unverified synthetic outputs.

Models were fine-tuned using PyTorch and HuggingFace’s Seq2SeqTrainer across a 5-fold cross-validation protocol with AdamW optimization and linear learning rate decay. To condition sequence-to-sequence architectures to operate specifically as dialect normalizers rather than generic machine translators, fine-tuning incorporates task-specific normalization conditioning tags and dialect-source identifiers (e.g., `<normalize>` task prefixes paired with dialect markers such as `<D1>`, `<D2>`, and `<D4>`). These special normalization tokens explicitly inform the encoder of the source dialect’s morpho-syntactic boundaries while enforcing strict target standardization (`<standard_mr>`) in the decoder, enabling precise sequence transduction into Standard Written Marathi.

Evaluation on the held-out 2,170-utterance IISc_RESPIN_test_mr benchmark confirms strong out-of-domain generalization across agricultural and financial text, with fine-tuned models reaching 76.50 BLEU for IndicBART and 73.48 BLEU for mT5-small. Importantly, models trained on synthetically expanded dialectal pairs maintained near-perfect normalization on Standard Written Marathi (97.39 BLEU and 1.37% WER for mT5-small), demonstrating that audited synthetic data expansion enhances dialectal coverage without causing catastrophic forgetting of standard language orthography.

Benchmarking pre-trained multilingual sequence-to-sequence architectures on the expanded benchmark establishes new state-of-the-art baselines: google/mT5-small (300M) achieved 69.51 BLEU and 84.58 chrF++ on the combined multi-dialect dataset, outperforming IndicBART (244M) by +12.39 BLEU. This performance margin is driven by key architectural advantages: mT5’s $3.9\times$ larger subword vocabulary (250K vs. 64K tokens) and relative positional biases accommodate agglutinative dialectal suffix merges far more effectively than IndicBART’s absolute sinusoidal encodings. Ultimately, this work provides a scalable, verified blueprint for expanding and benchmarking low-resource Indic sub-dialects without reliance on manual human annotation.

5.1 Limitations

This work has several limitations that should be acknowledged:

1. Synthetic data generation relies on LLM inference pipelines (Gemma-4 via local/hosted endpoints), introducing computational resource and latency considerations for large-scale replication.
2. While we benchmarked two encoder-decoder architectures (mT5-small and IndicBART), decoder-only LLMs (e.g., LLaMA-3, Gemma-4) have not been evaluated for direct zero-shot or fine-tuned text normalization.
3. Evaluation is currently focused on intrinsic text normalization metrics (BLEU, chrF++, WER); downstream task evaluations (e.g., ASR post-processing, translation) remain for future work.
4. Synthetic augmentation exhibited a slight negative yield on Varhadi (−0.84 BLEU for mT5-small), suggesting that for already high-performing dialects, over-generalized suffix rules in synthetic data may slightly dilute precision—highlighting the need for targeted augmentation strategies.

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